

ECE 3337: Signals & Systems Analysis

Lecture 21: Fourier Analysis Techniques

**Reference: Chapter 5
(Section 5.9 – 5.10)**

Please turn off your gadgets now



Announcements

- Dr. Roysam will be on travel week of April 20th
- Classes will be held online by Dr. Reddy
- The Zoom link is below:

<https://uh-edu-cougarnet.zoom.us/j/89206792357?pwd=MdV6UBmjZ2jnmaXramZRQmpLWHEdNw.1>

Meeting ID: 892 0679 2357

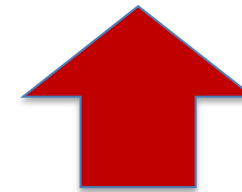
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- Make sure to pickup your graded exams today
–*Deadline for Regrade requests is Friday 5pm*



Recap: Three Forms of Fourier Series

Cosine/Sine	Amplitude/Phase	Complex Exponential
$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$	$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$	$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$
$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt$	$c_n e^{j\phi_n} = a_n - jb_n$	$\mathbf{x}_n = \mathbf{x}_n e^{j\phi_n}; \mathbf{x}_{-n} = \mathbf{x}_n^*; \phi_{-n} = -\phi_n$
$a_n = \frac{2}{T_0} \int_0^{T_0} x(t) \cos n\omega_0 t dt$	$c_n = \sqrt{a_n^2 + b_n^2}$	$ \mathbf{x}_n = c_n/2; x_0 = c_0$
$b_n = \frac{2}{T_0} \int_0^{T_0} x(t) \sin n\omega_0 t dt$	$\phi_n = \begin{cases} -\tan^{-1}(b_n/a_n), & a_n > 0 \\ \pi - \tan^{-1}(b_n/a_n), & a_n < 0 \end{cases}$	$\mathbf{x}_n = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jn\omega_0 t} dt$
$a_0 = c_0 = x_0; a_n = c_n \cos \phi_n; b_n = -c_n \sin \phi_n; \mathbf{x}_n = \frac{1}{2}(a_n - jb_n)$		





Recap: Exponential Representation

$$\begin{aligned}x(t) &= a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right] \\&= a_0 + \sum_{n=1}^{\infty} \left[\frac{(a_n - jb_n)}{2} e^{jn\omega_0 t} + \frac{(a_n + jb_n)}{2} e^{-jn\omega_0 t} \right] \\&= a_0 + \sum_{n=1}^{\infty} \left[\mathbf{x}_n e^{jn\omega_0 t} + \mathbf{x}_{-n} e^{-jn\omega_0 t} \right] = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}\end{aligned}$$

$$\mathbf{x}_n = \frac{(a_n - jb_n)}{2}$$

$$\mathbf{x}_0 = a_0$$

$$\mathbf{x}_{-n} = \frac{(a_n + jb_n)}{2} = \mathbf{x}_n^*$$



Today

1. We will develop an understanding of signal power in the context of the Fourier series
2. We will examine the Fourier series in the extreme case of $T_0 \rightarrow \infty$



Recap: Signal Power & Energy

The notions of power and energy apply to signals, regardless of the underlying physical reality

$$\text{Instantaneous Power} = p(t) = |x(t)|^2 = x(t)x^*(t)$$

Total Energy E Expended from t_1 to t_2 :

$$E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} |x(t)|^2 dt = \int_{t_1}^{t_2} x(t)x^*(t) dt$$

$$\text{Average Power: } P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} p(t) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} |x(t)|^2 dt$$



Special case: Periodic Signals

For a periodic signal, the average power is the same as the average power over a single time period

$$P_{av} = \frac{1}{T_0} \int_{-T_0/2}^{+T_0/2} |x(t)|^2 dt = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt$$

We only care that the integration period = T_0



Average Power of a Sinusoid

$$x_1(t) = A \cos(n\omega_0 t + \phi), \quad \omega_0 \neq 0, \quad n > 0$$

We only need to do the calculation over one period since it is periodic

$$\begin{aligned} P_{x1} &= \frac{1}{T_0} \int_0^{T_0} |A \cos(n\omega_0 t + \phi)|^2 dt \\ &= \frac{A^2}{T_0} \int_0^{T_0} \cos^2(n\omega_0 t + \phi) dt \\ &= \frac{A^2}{T_0} \int_0^{T_0} \left[\frac{1}{2} (1 + \cos 2(n\omega_0 t + \phi)) \right] dt = \frac{A^2}{2} \end{aligned}$$



This Formula is Commonly Used

- Often, signal-related quantities are specified in terms of their “root mean squared (RMS)” equivalent
 - It is a way to relate the quantity to an equivalent DC value
 - For example, a one Ampere RMS sinusoidal current would produce the same heating in a resistor as a 1 Ampere DC current
- Example: US wall power outlets have a sinusoidal voltage signal of 120V RMS (root mean square) at a frequency of 60Hz

$$\frac{A}{\sqrt{2}} = 120, \Rightarrow A = 120\sqrt{2} = 169.7 \text{ (peak voltage)}$$

Shocking



Power of a Complex Exponential

$$x_2(t) = Ae^{j(\omega_0 t + \phi)}$$

$$P_{x_2} = \frac{1}{T_0} \int_0^{T_0} |x_2(t)|^2 dt = \frac{1}{T_0} \int_0^{T_0} x_2(t) x_2^*(t) dt$$

$$= \frac{1}{T_0} \int_0^{T_0} Ae^{j(\omega_0 t + \phi)} \times Ae^{-j(\omega_0 t + \phi)} dt = A^2$$



Power of a Sum of Sinusoids

We want to calculate the average power for a general signal represented by a Fourier Series.

As a first step, consider just two signals:

$$x(t) = c_n \cos(n\omega_0 t + \phi_n)$$

$$y(t) = c_m \cos(m\omega_0 t + \phi_m), n \neq m$$

From our calculation on the previous slide, it is easy to see that:

$$P_x = \frac{1}{T_0} \int_0^{T_0} c_n^2 \cos^2(n\omega_0 t + \phi_n) dt = \frac{c_n^2}{2}$$



Power of a Sum of Sinusoids

Similarly:

$$P_y = \frac{1}{T_0} \int_0^{T_0} c_m^2 \cos^2(m\omega_0 t + \phi_m) dt = \frac{c_m^2}{2}$$

Let's see what happens when we add two signals $x(t)$ and $y(t)$:

$$\begin{aligned} P_{x+y} &= \frac{1}{T_0} \int_0^{T_0} |x(t) + y(t)|^2 dt = \frac{1}{T_0} \int_0^{T_0} [x(t) + y(t)][x^*(t) + y^*(t)] dt \\ &= \underbrace{\frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt + \frac{1}{T_0} \int_0^{T_0} |y(t)|^2 dt}_{\text{Green bracket}} + \underbrace{\frac{1}{T_0} \int_0^{T_0} [x(t)y^*(t) + x^*(t)y(t)] dt}_{\text{Red bracket}} \end{aligned}$$



Sum of Sinusoids (cont.)

$$P_{x+y} = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt + \frac{1}{T_0} \int_0^{T_0} |y(t)|^2 dt + \frac{1}{T_0} \int_0^{T_0} [x(t)y^*(t) + x^*(t)y(t)] dt$$

$$\frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = \frac{c_n^2}{2}$$

$$\frac{1}{T_0} \int_0^{T_0} |y(t)|^2 dt = \frac{c_m^2}{2}$$

$$\frac{1}{T_0} \int_0^{T_0} [x(t)y^*(t) + x^*(t)y(t)] dt$$

$$= \frac{1}{T_0} \int_0^{T_0} 2c_n c_m \cos(n\omega_0 t + \phi_n) \cos(m\omega_0 t + \phi_m) dt = 0$$

$$P_{x+y} = \frac{c_n^2}{2} + \frac{c_m^2}{2}$$

**THIS IS A COOL AND IMPORTANT
RESULT SINCE WE CAN GENERALIZE
IT TO THE ENTIRE FOURIER SERIES**



Recap: Three Forms of Fourier Series

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)].$$

(sine/cosine representation) (5.27)

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n),$$

(amplitude/phase representation)

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t},$$

(exponential representation)



3 Versions of Power Formula

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)/2, \quad (5.58a)$$

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = c_0^2 + \sum_{n=1}^{\infty} c_n^2/2, \quad (5.58b)$$

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |\mathbf{x}_n|^2. \quad (5.58c)$$



THIS IS CALLED PARSEVAL'S THEOREM (1799 in Honor of Marc-Antoine Parseval):

- **THE POWER OF A SUM OF SINUSOIDS IS EQUAL TO THE SUM OF THE POWERS OF ITS FOURIER COMPONENTS**
- **BASICALLY, the FOURIER REPRESENTATION DOES NOT LOSE ANY ENERGY OF THE SIGNAL**



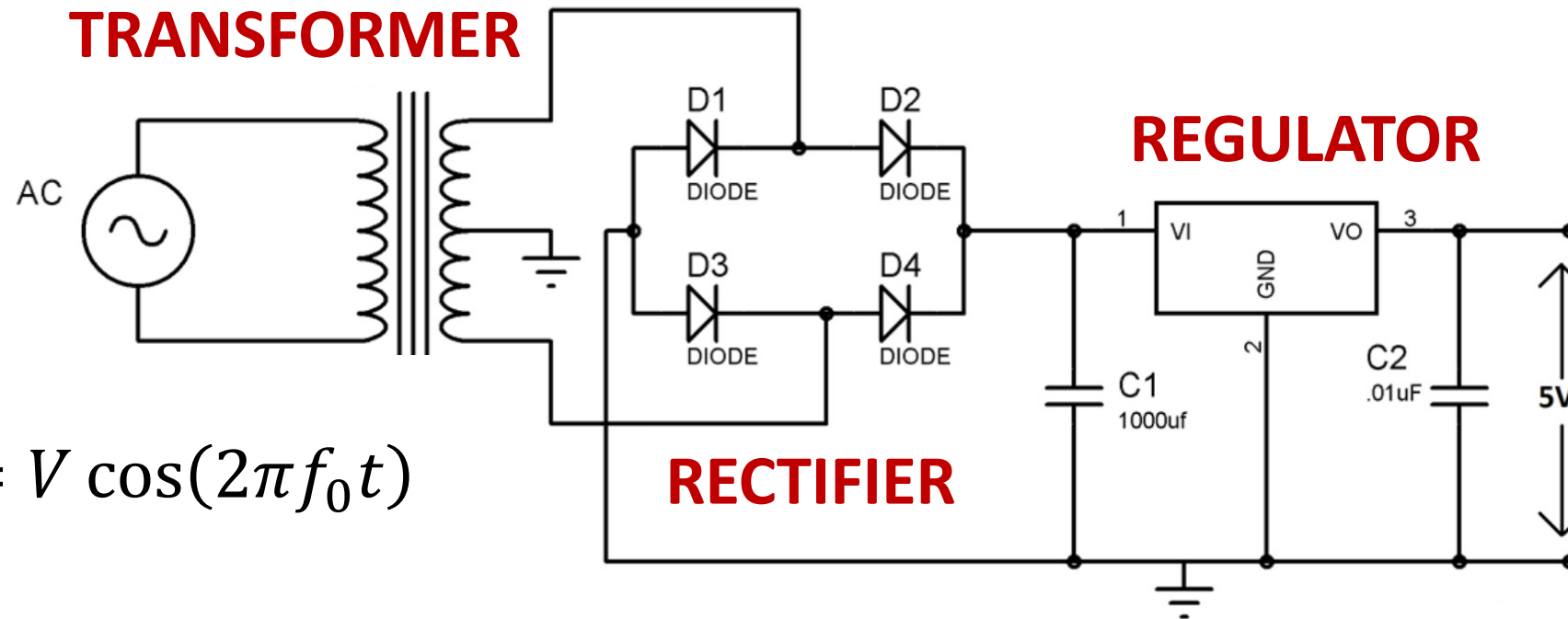
AC and DC Power

- The power associated with the zero frequency terms (a_0 , c_0 , or $|x_0|$) is called the **dc power**
- The remaining power associated with the harmonics is referred to as the **ac power**
- The **ac power fraction** is the ratio of the *ac* power and the sum of the *ac* power and *dc* power
 - This is an important issue for power supplies, e.g., a cell-phone charger
 - Ideally, the phone needs a steady 5V dc, but this must be derived from the mains voltage

$$v_{ac}(t) = 170 \cos(2\pi f_0 t), \text{ where } f_0 = 60 \text{ Hz}$$



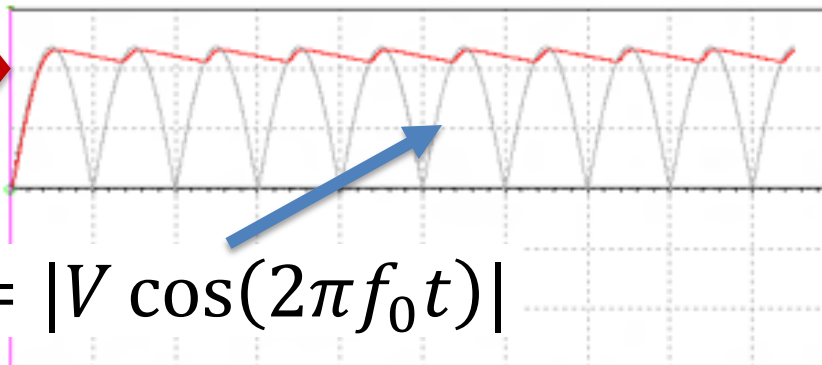
Cell-phone Charger: The Ripple Problem



**OUTPUT WITH
MINIMUM AC
CONTENT**

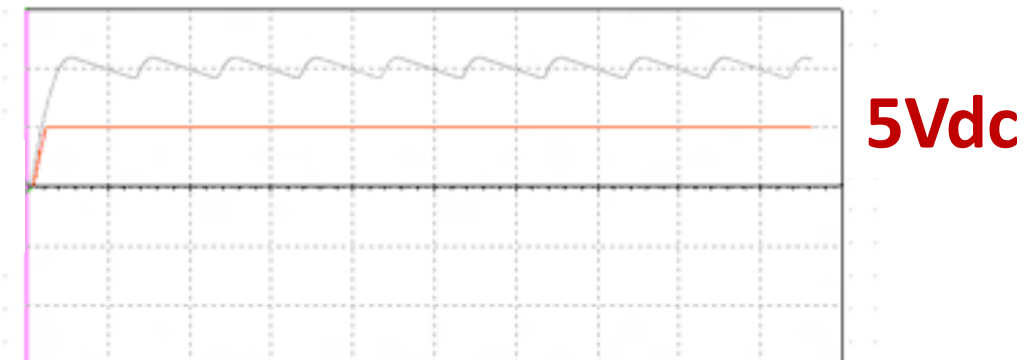
$$v_{ac}(t) = V \cos(2\pi f_0 t)$$

RIPPLE →



$$v_{rect}(t) = |V \cos(2\pi f_0 t)|$$

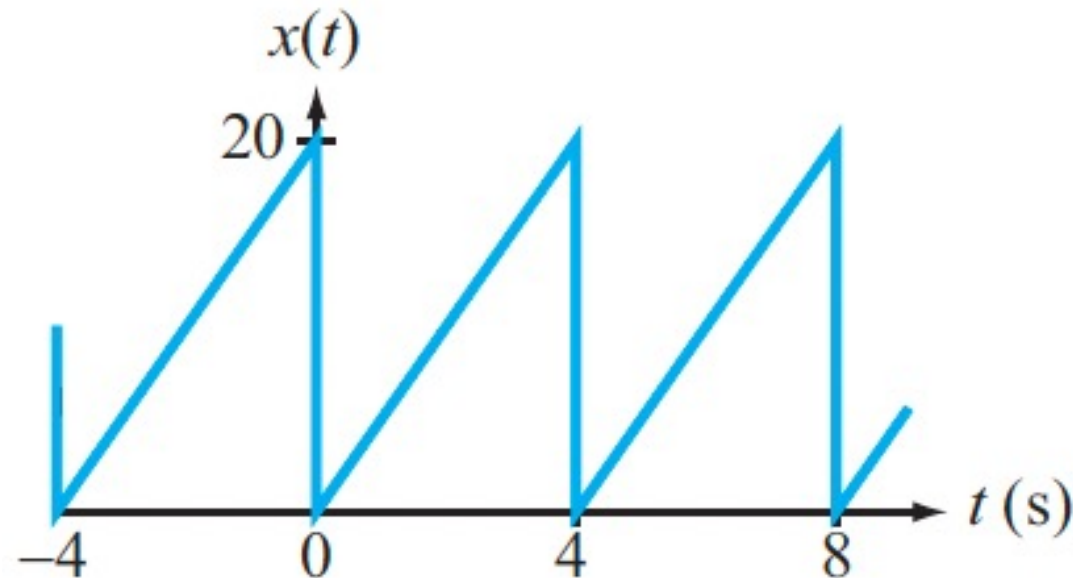
REGULATOR →





Example

Consider the sawtooth waveform:



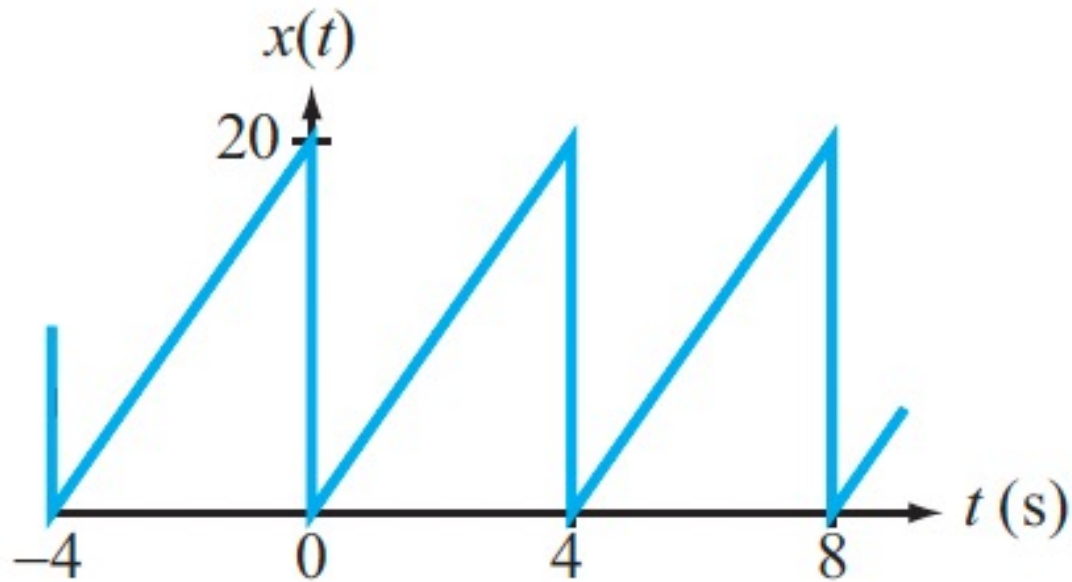
We can calculate the average power in two equivalent ways:

1. Directly integrate over one period
2. Calculate from the Fourier Series expansion



Direct Integration

Average the power over one period:



$$T_0 = 4$$

$$x(t) = 5t$$

(a ramp within each period)

$$P_x = \frac{1}{4} \int_0^4 (5t)^2 dt = \frac{400}{3}$$



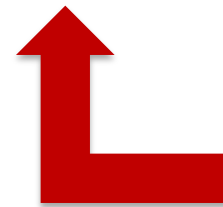
From the Fourier Series

We had calculated the Fourier Series for this signal earlier:

$$x(t) = 10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2}t\right)$$

$$\Rightarrow P_x = 10^2 + \sum_{n=1}^{\infty} \left(\frac{20}{\pi}\right)^2 \times \frac{1}{2n^2}$$

$$= P_x = 100 + \frac{200}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$



The Basel Problem: Genius Euler Strikes Again

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

- There is fascinating history behind this formula.
- It was proposed by Italian mathematician Pietro Mengoli in 1650, but he could not prove it
- It was famously proved in 1735 by Leonhard Euler, and then named in honor of his hometown of Basel in Switzerland





From the Fourier Series

We had calculated the Fourier Series earlier:

$$x(t) = 10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2}t\right)$$

$$\Rightarrow P_x = 10^2 + \sum_{n=1}^{\infty} \left(\frac{20}{\pi}\right)^2 \times \frac{1}{2n^2}$$

$$= P_x = 100 + \frac{200}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$= \frac{400}{3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

SAME ANSWER, SO PARSEVAL'S THEOREM WORKS



What does this tell us?

The Bottom Line:

The Fourier series representation of a periodic signal accounts for all of the signal power

- **Again, this works for all signals, whether or not they represent any physical quantities (e.g., voltage or current)**



Non-periodic Signals

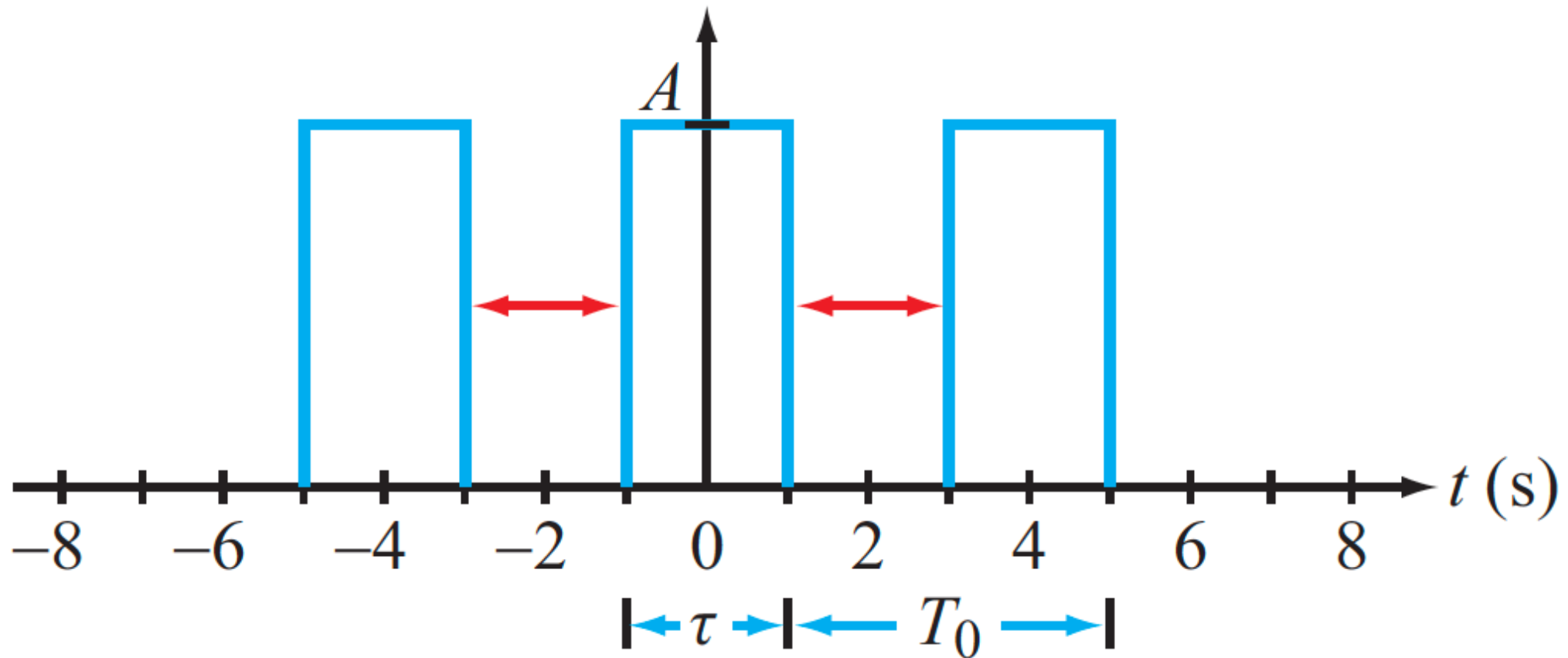
A non-periodic signal can be thought of as the extreme case of a periodic signal, when the time period approaches infinity

- **Intuitively, as the time period approaches infinity, the signal pattern cannot repeat again**
- **So, we could find the Fourier series representation of a periodic signal, and then ask what happens as:**

$$T_0 \rightarrow \infty$$



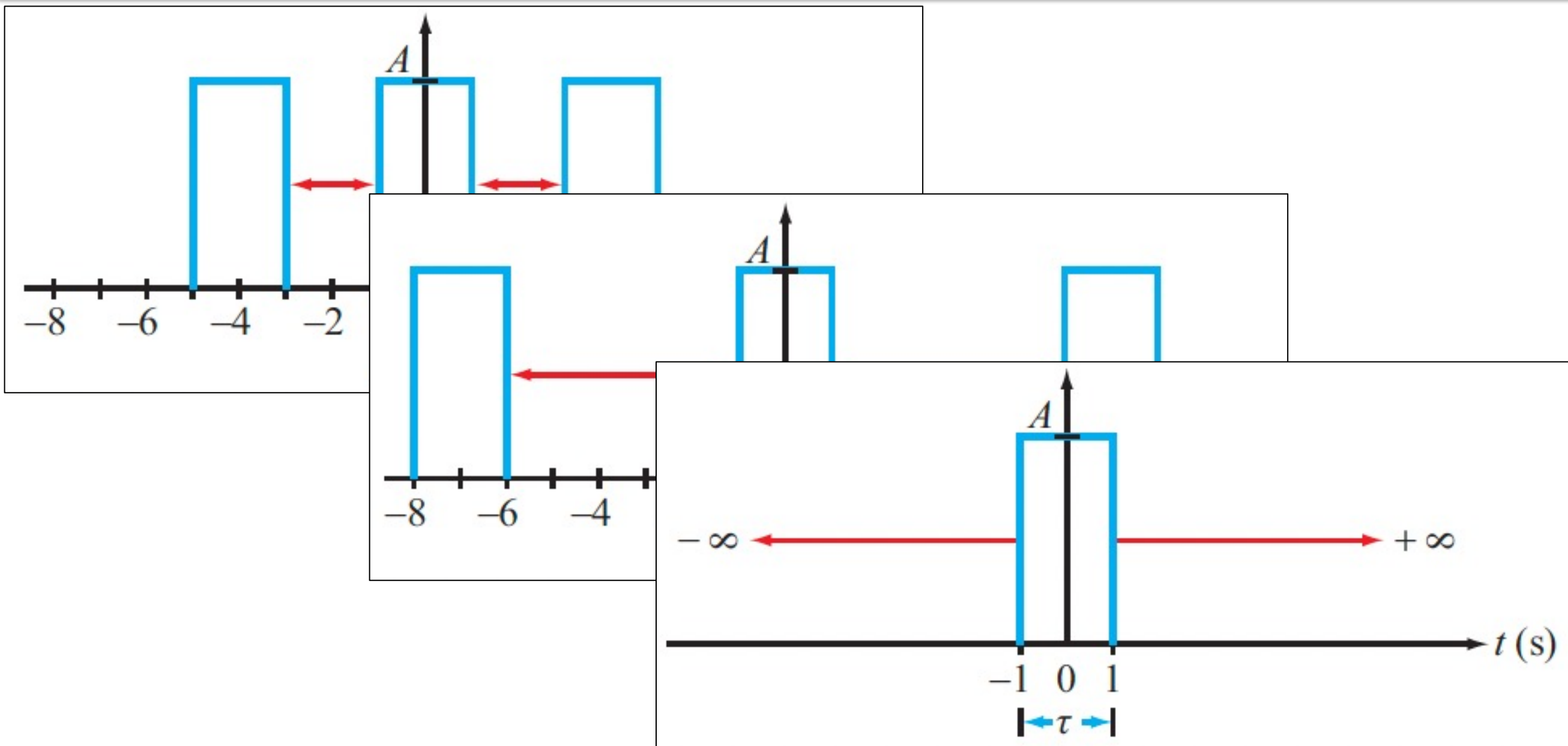
Pulse Train Example



OVER ONE PERIOD:
$$x(t) = \begin{cases} A & -\tau / 2 \leq t \leq \tau / 2 \\ 0 & \text{else} \end{cases}$$



Pushing the Period Towards Infinity





Fourier Series of the Pulse Train

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t},$$

(exponential representation)

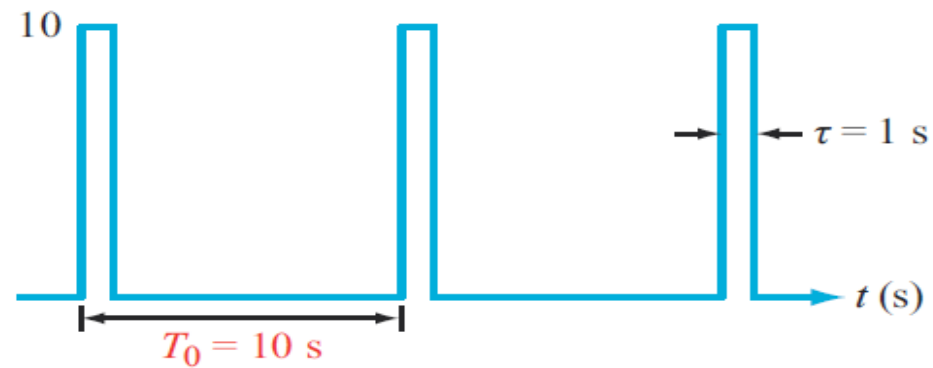
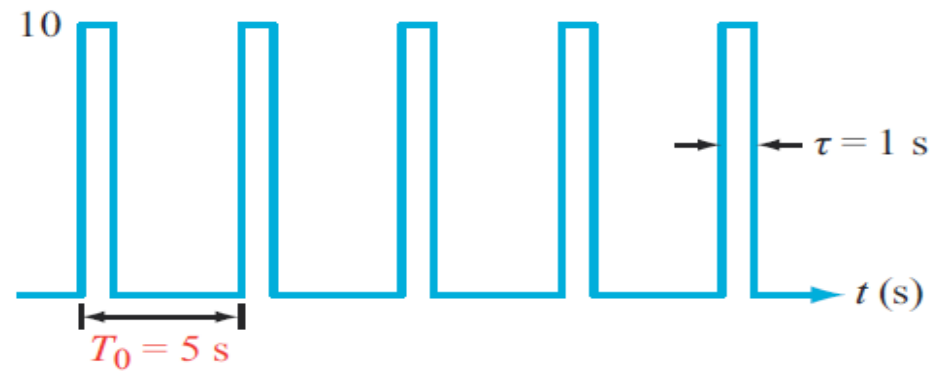
$$\begin{aligned} \mathbf{x}_n &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} A e^{-jn\omega_0 t} dt \\ &= \left(\frac{A e^{-jn\omega_0 t}}{-jn\omega_0 T_0} \right) \bigg|_{-\tau/2}^{\tau/2} = \frac{2A}{n\omega_0 T_0} \left[\frac{e^{jn\omega_0 \tau/2} - e^{-jn\omega_0 \tau/2}}{2j} \right] = \frac{2A}{n\omega_0 T_0} \sin(n\omega_0 \tau / 2) \end{aligned}$$

LET US PLOT THIS FOR INCREASING TIME PERIODS

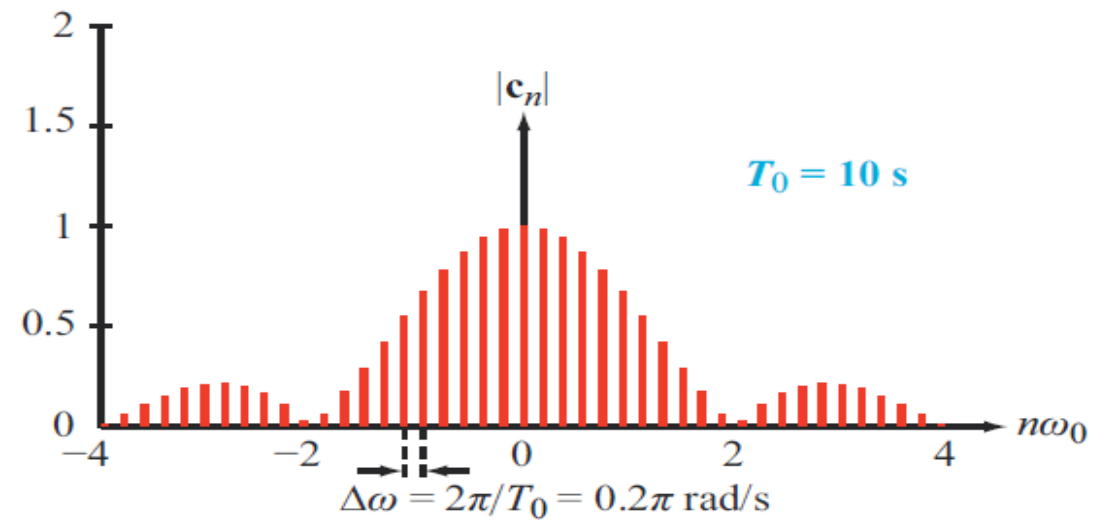
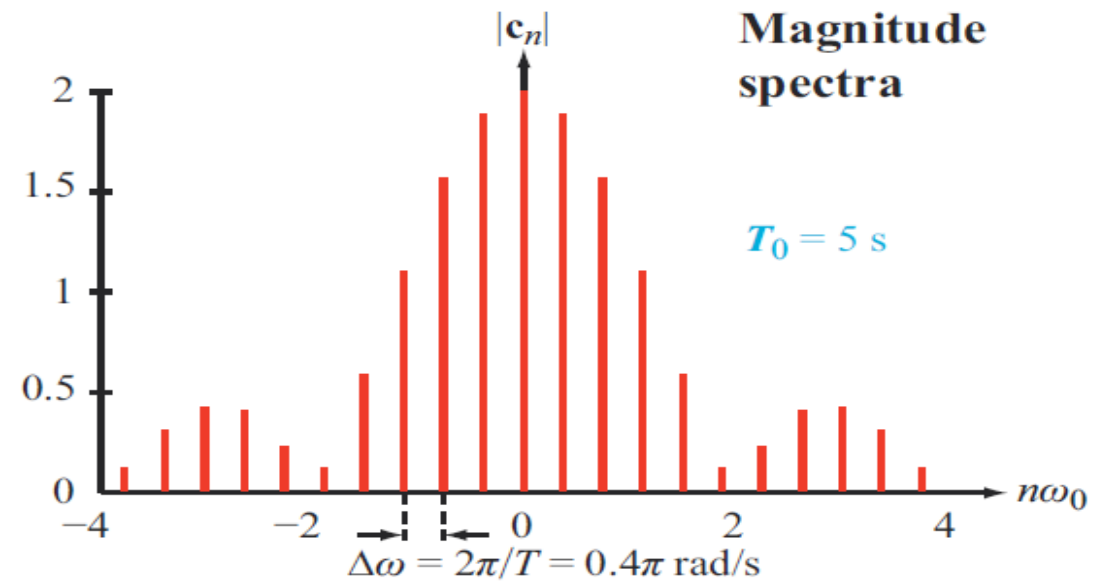


Increasing the Period

Waveforms $x(t)$



Magnitude spectra





Observations

The spacing between successive harmonics n and $(n + 1)$ is given by:

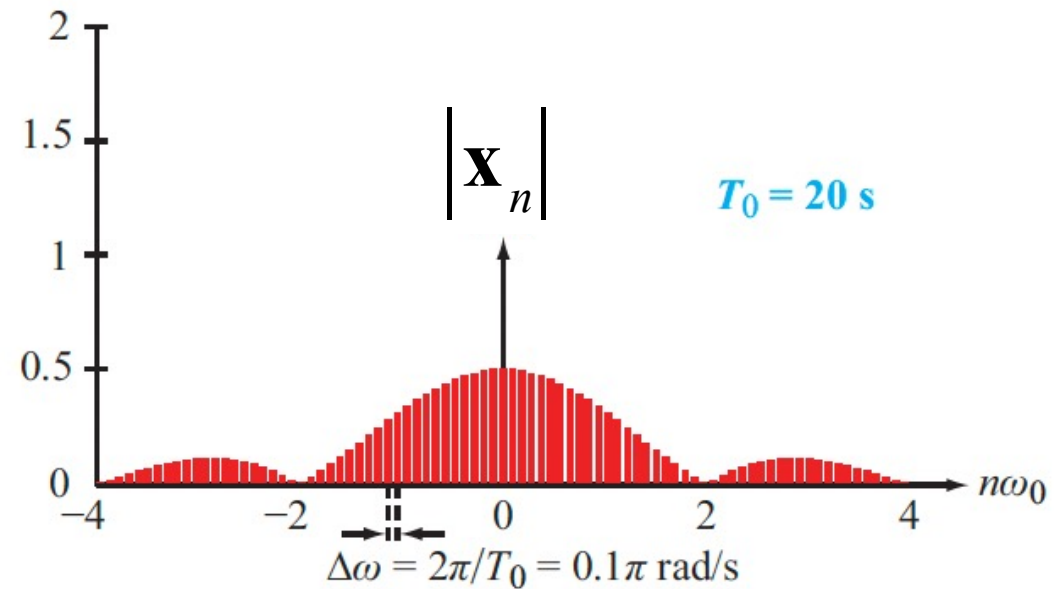
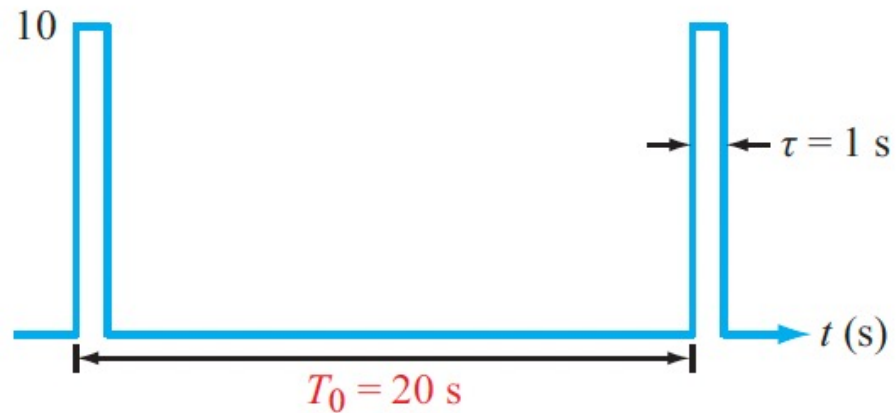
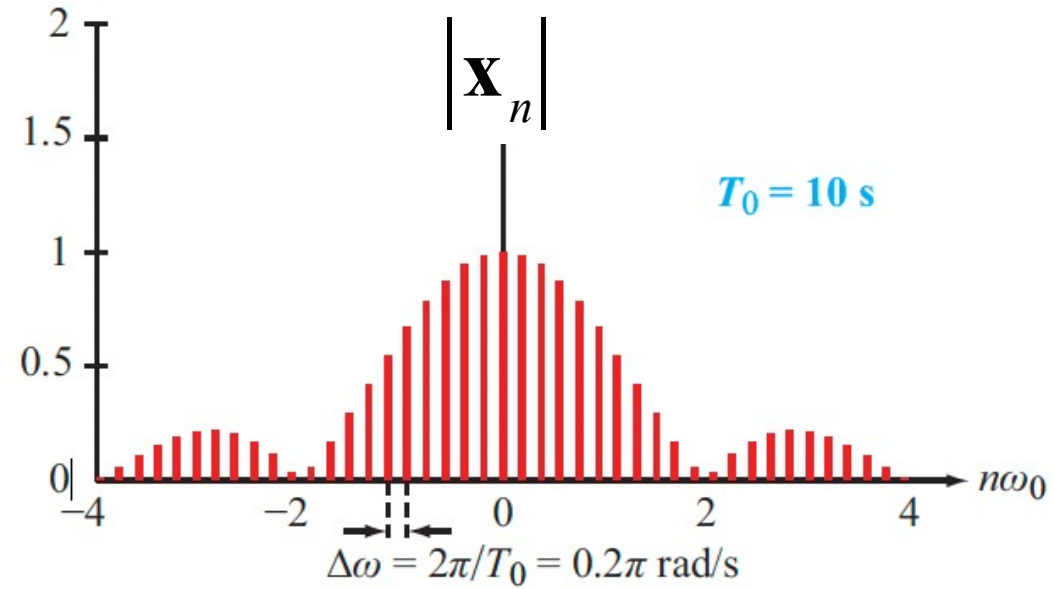
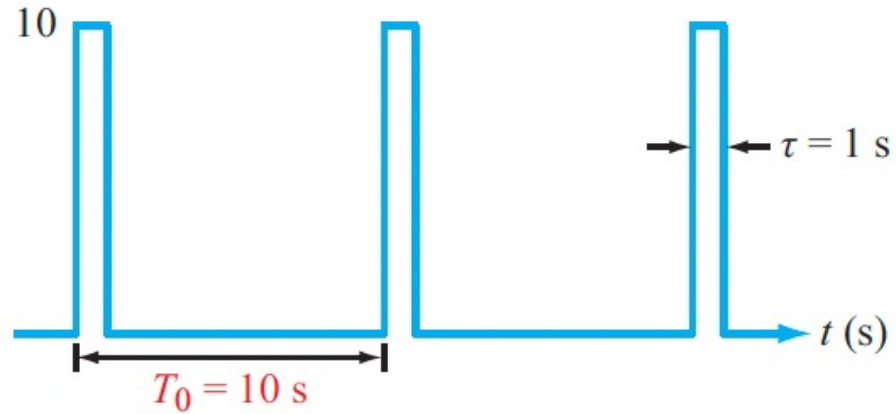
$$\Delta\omega = (n + 1)\omega_0 - n\omega_0 = \omega_0 = \frac{2\pi}{T_0}$$

$$\text{As } T_0 \rightarrow \infty, \quad \Delta\omega \rightarrow 0$$

The spectral lines become more and more dense with increasing T_0 , and in the limit, approach a continuum



Increasing the Period





Observations

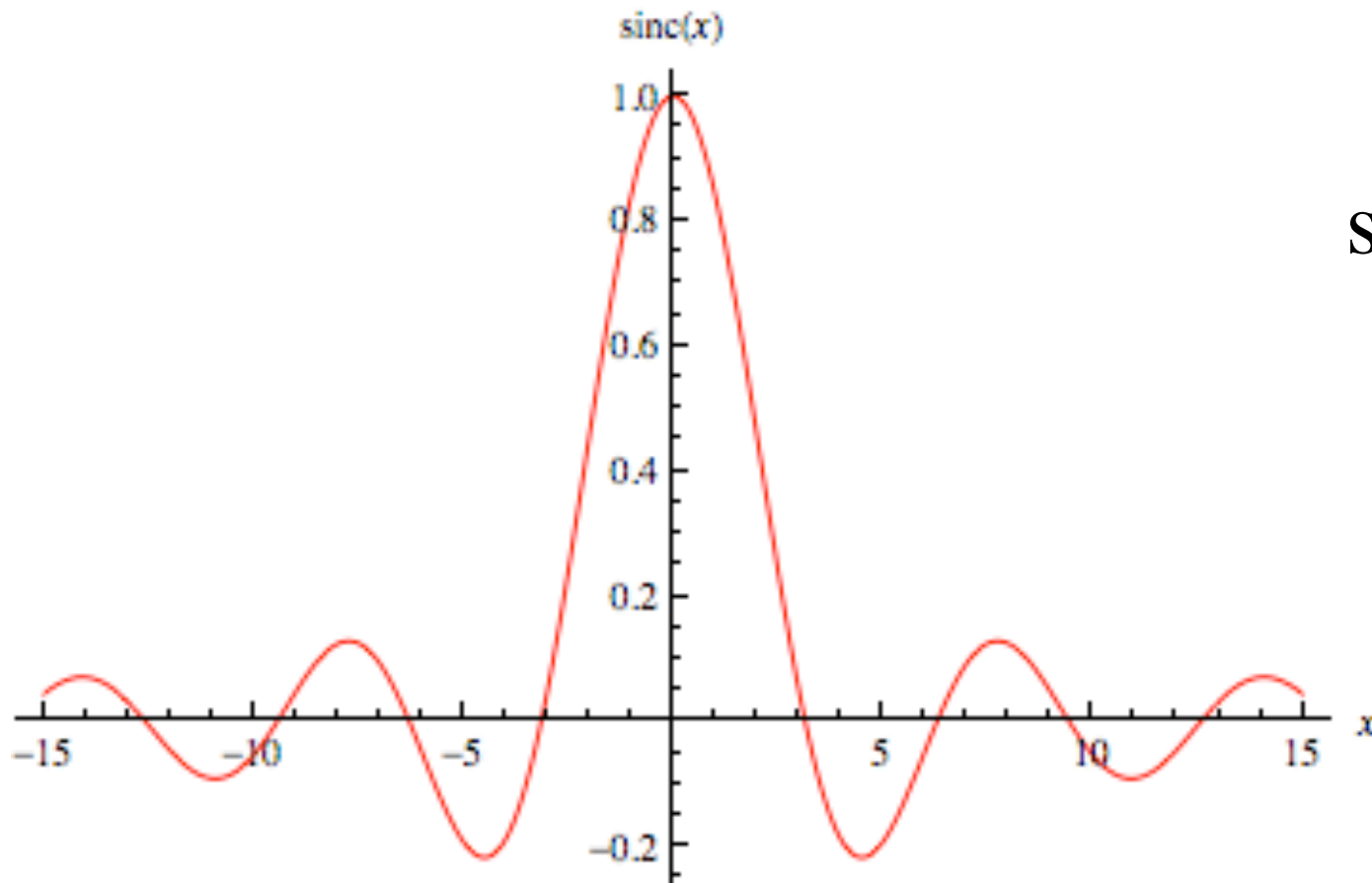
However, as we increase the period T_0 , the overall shape of the spectrum remains the same

$$\begin{aligned} \mathbf{x}_n &= \frac{2A}{n\omega_0 T_0} \sin(n\omega_0 \tau / 2) \\ &= \frac{A\tau}{T_0} \times \frac{\sin(n\omega_0 \tau / 2)}{(n\omega_0 \tau / 2)} = \frac{A\tau}{T_0} \underline{\text{sinc}(n\omega_0 \tau / 2)} \end{aligned}$$

THE SINC FUNCTION APPEARS OFTEN IN SIGNAL ANALYSIS
SINCE PULSE TRAINS ARE SO COMMON



The Sinc Function

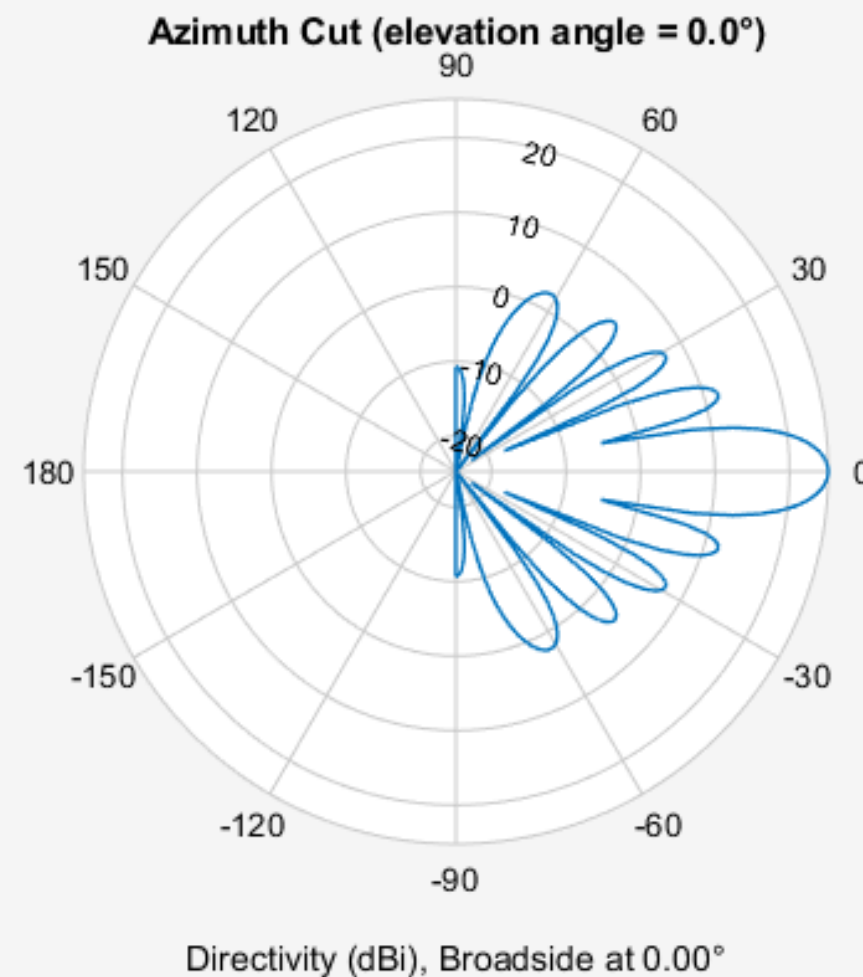
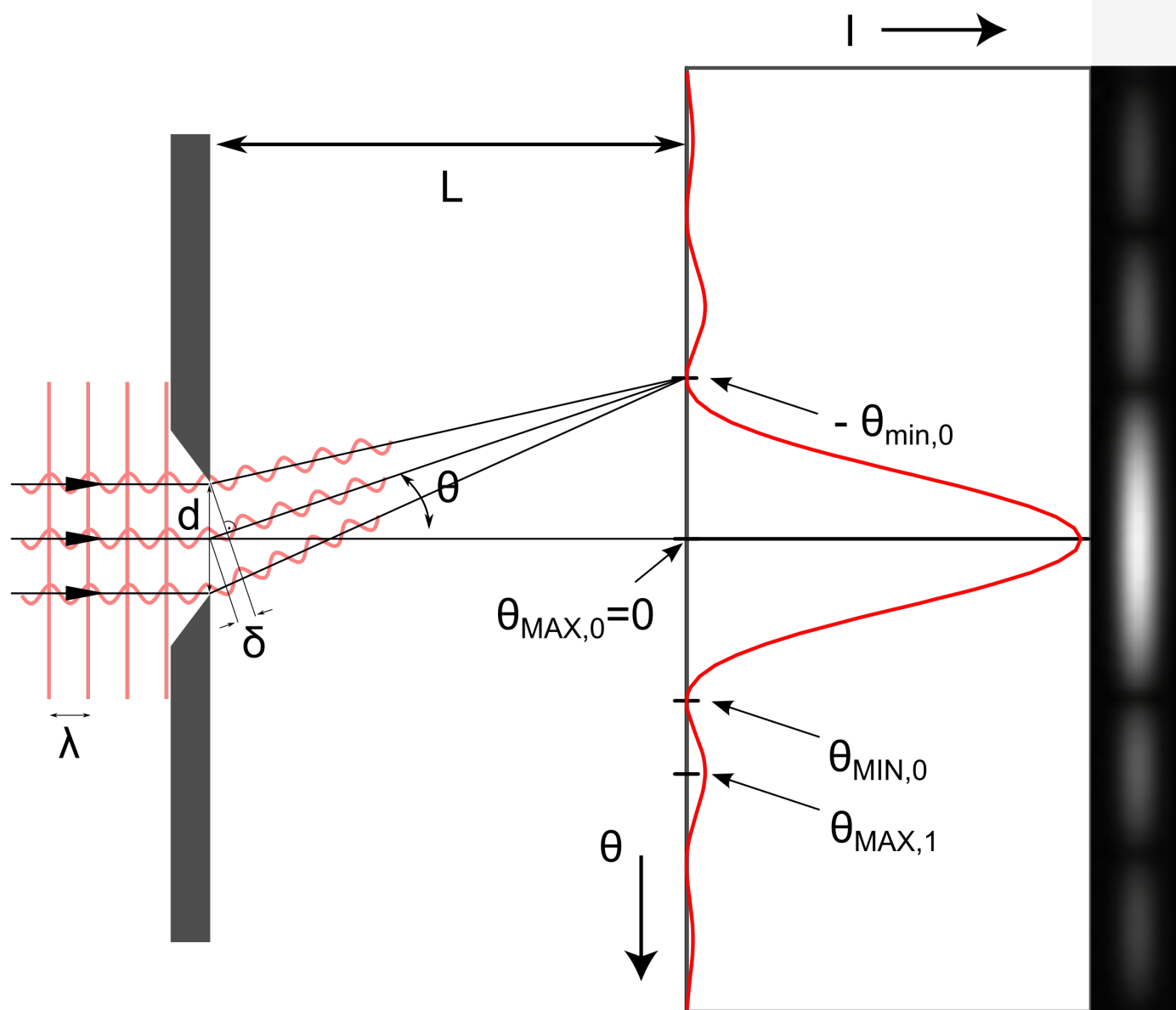


$$\text{sinc}(x) = \begin{cases} 1 & \text{for } x = 0 \\ \frac{\sin(x)}{x} & \text{otherwise} \end{cases}$$

$$\lim_{x \rightarrow 0} \left\{ \frac{\sin(x)}{x} \right\} = 1$$

The name “sinc” is derived from “sine cardinal”
This function appears so often in that it deserves its own name

Remember Single-slit Diffraction or Antennas?



**Antenna Radiation Pattern
(MATLAB function)**



Introducing the Fourier Transform

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t},$$

(exponential representation)

$$\mathbf{x}_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(\tau) e^{-jn\omega_0 \tau} d\tau \Rightarrow x(t) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(\tau) e^{-jn\omega_0 \tau} d\tau \right] e^{jn\omega_0 t}$$

Substitute $\Delta\omega = \frac{2\pi}{T_0}$, or $\frac{1}{T_0} = \frac{\Delta\omega}{2\pi}$



Introducing the Fourier Transform

$$x(t) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(\tau) e^{-jn\omega_0\tau} d\tau \right] e^{jn\omega_0 t}$$

Substitute $\frac{1}{T_0} = \frac{\Delta\omega}{2\pi}$

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-T_0/2}^{T_0/2} x(\tau) e^{-jn\omega_0\tau} d\tau \right] e^{jn\omega_0 t} \Delta\omega$$

In the limit, as $T_0 \rightarrow \infty$, the summation becomes an integral:

$$x(t) = \int_{-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau \right] e^{j\omega t} d\omega$$



The Fourier Transform

$$\hat{\mathbf{X}}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau$$

$$x(t) = \mathcal{F}^{-1}[\hat{\mathbf{X}}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\mathbf{X}}(\omega) e^{j\omega t} d\omega$$

$$x(t) \leftrightarrow \hat{\mathbf{X}}(\omega)$$

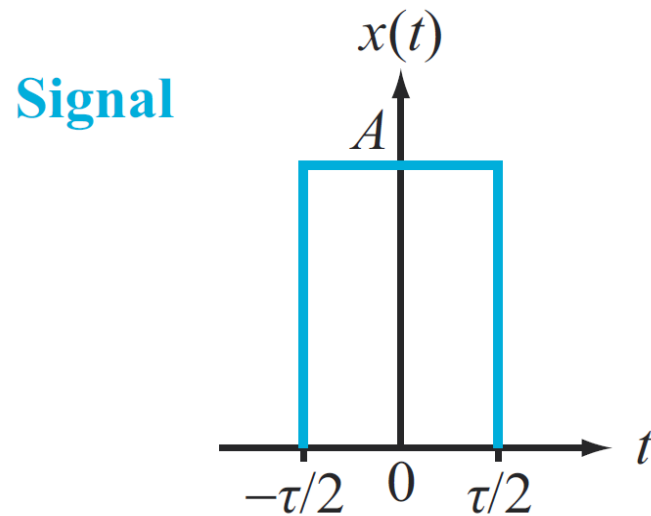
The hat ^ distinguishes this from the Laplace Transform
WE WILL EXPLORE THIS TRANSFORM NEXT CLASS



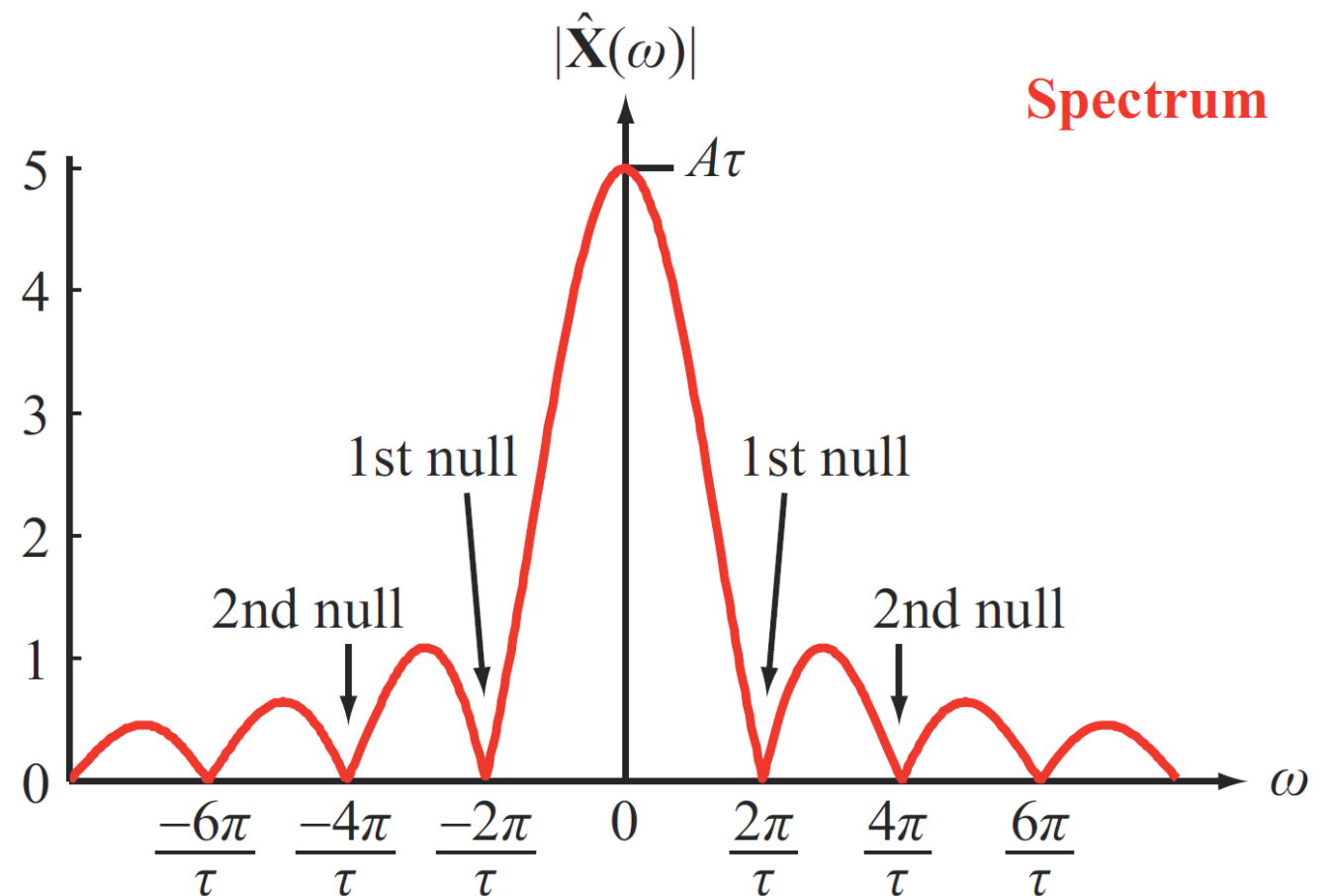
The Spectrum of a Pulse

**A super
important signal**

$$x(t) = A \operatorname{rect}(t/\tau)$$



$$\hat{\mathbf{X}}(\omega) = A\tau \frac{\sin(\omega\tau/2)}{(\omega\tau/2)} = A\tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$

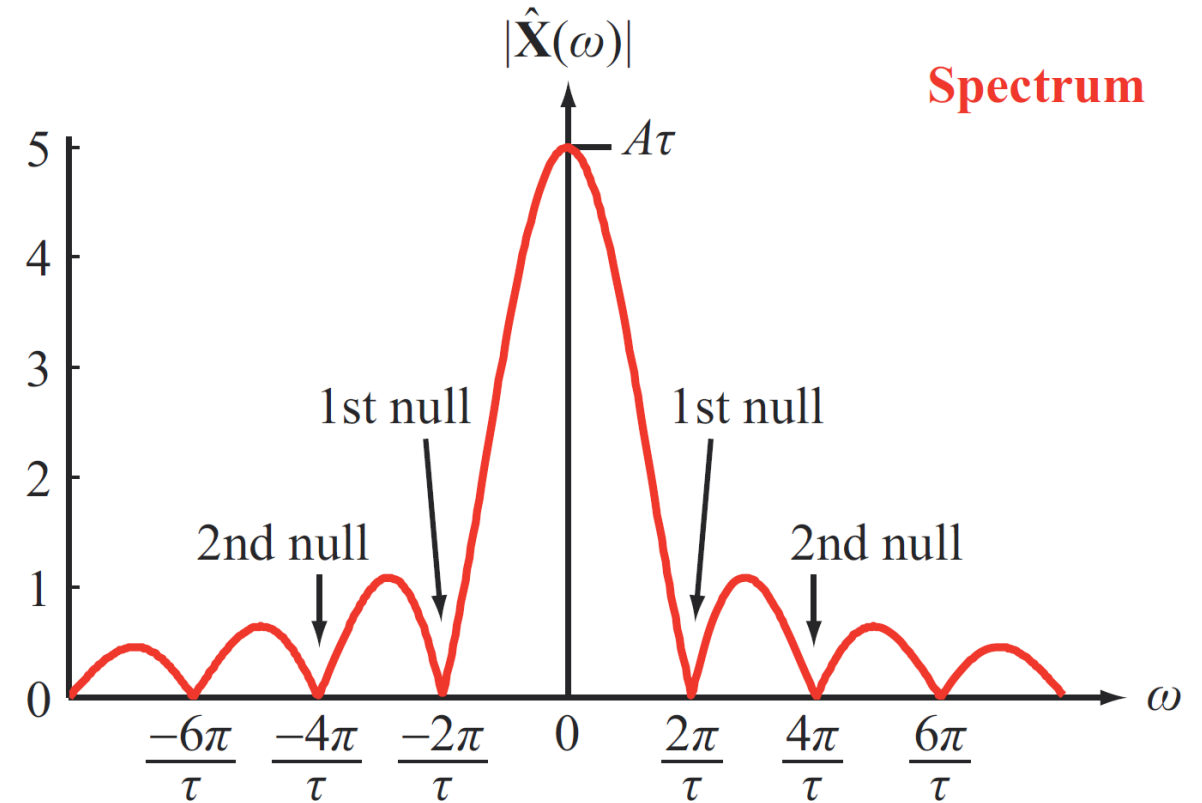
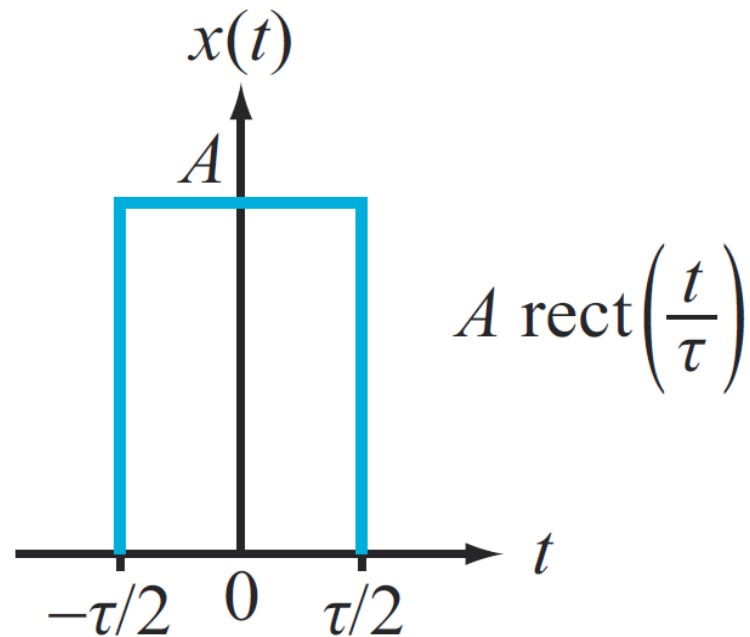




Pulse-width and Bandwidth

Let us consider what happens when we vary the pulse width τ

Signal

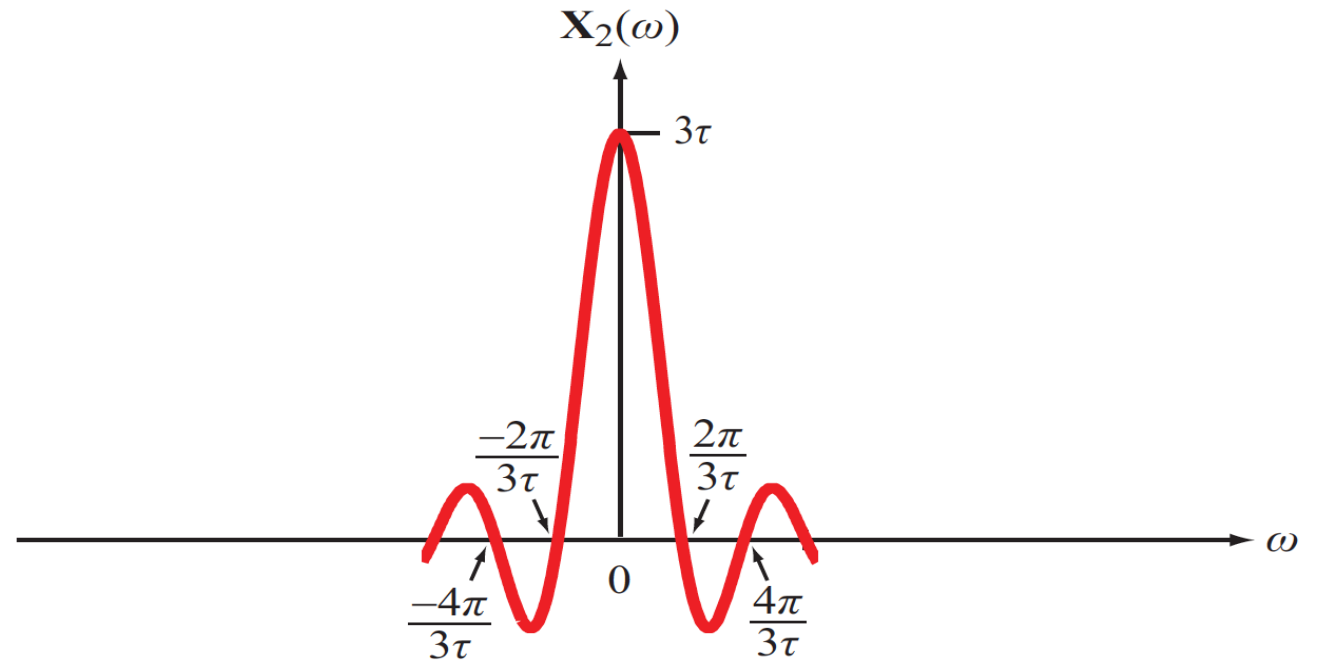
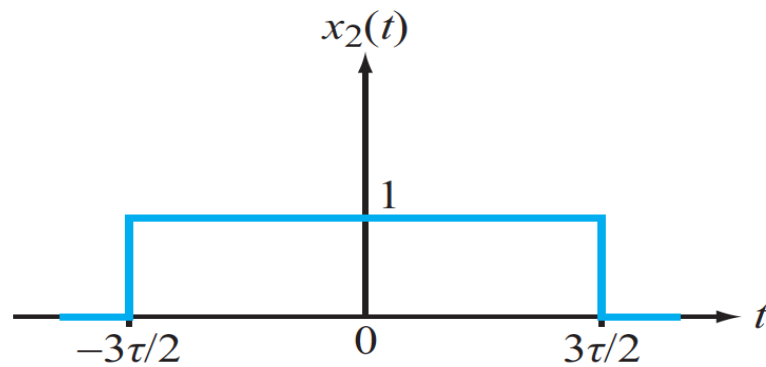
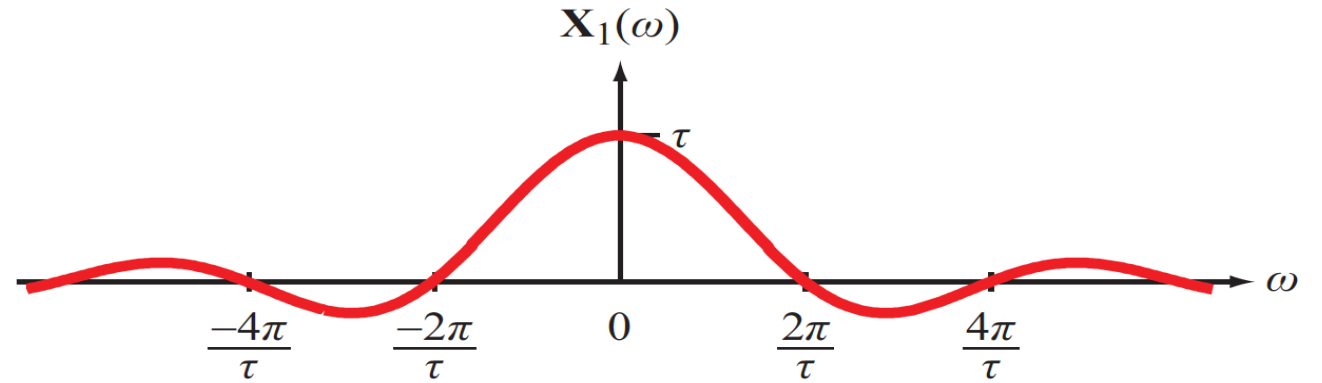
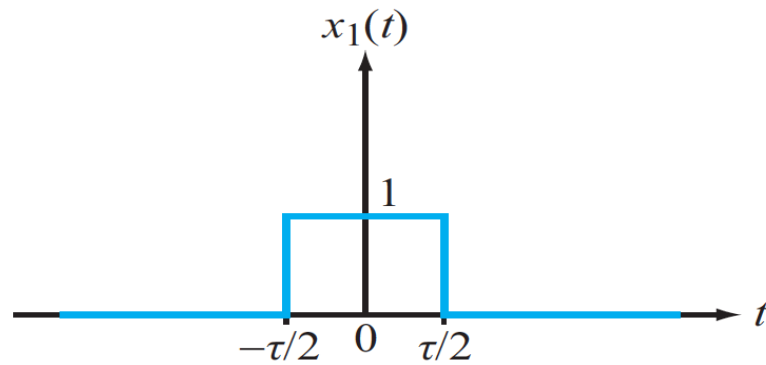


A narrower pulse corresponds to a wider spectrum, and *vice versa*

Can you think of an interesting extreme case?

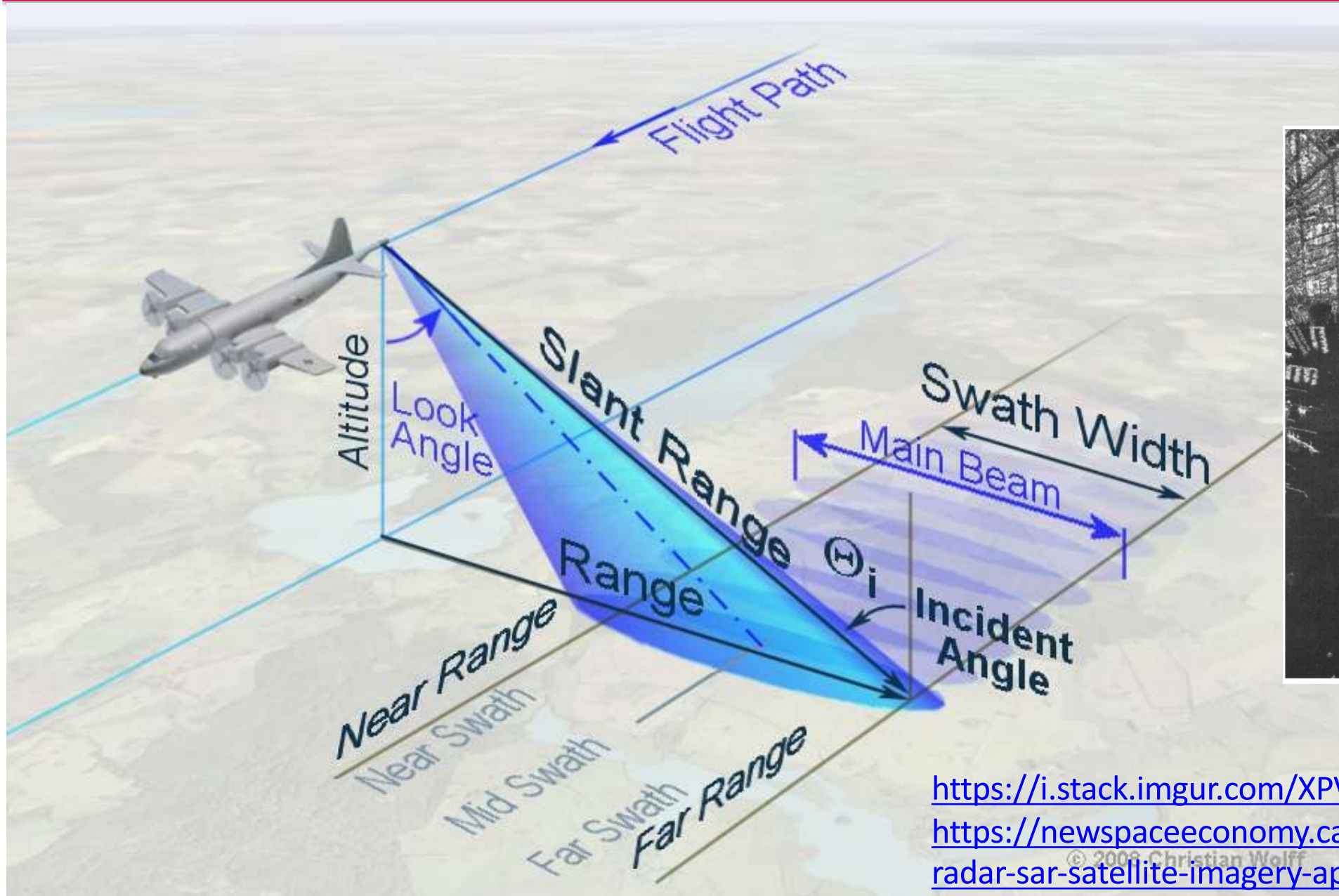


Pulse-width and Bandwidth





Example: Synthetic Aperture Radar



<https://i.stack.imgur.com/XPVpz.jpg>

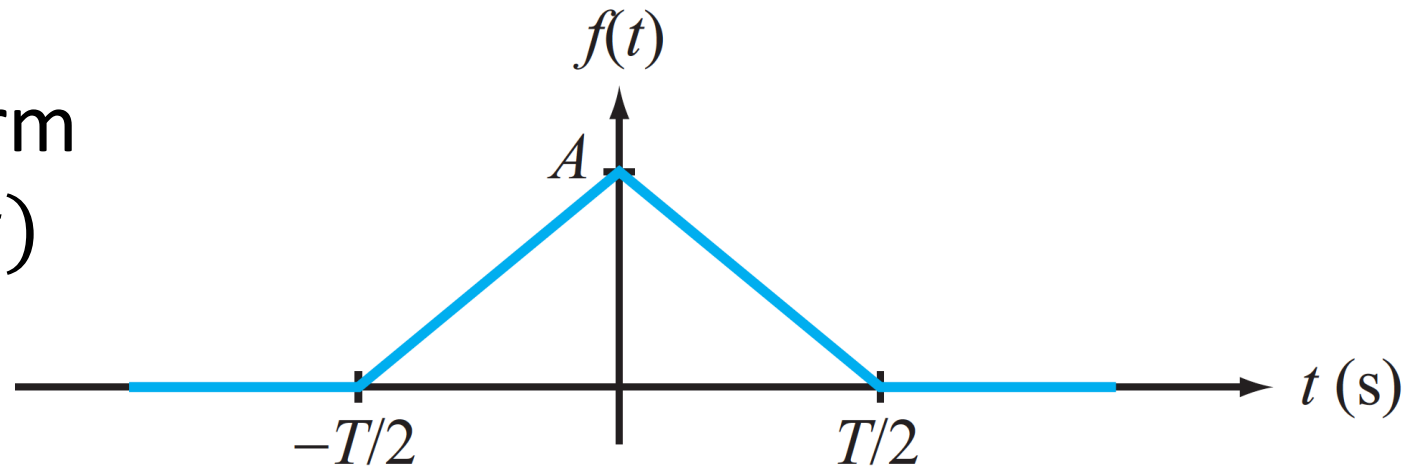
<https://newspaceconomy.ca/2023/03/19/synthetic-aperture-radar-sar-satellite-imagery-applications/>

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Try This Now

Show that the Fourier Transform $F(\omega)$ for the triangle pulse $f(t)$ is given by:



$$F(\omega) = \frac{4A}{T\omega^2} \left[1 - \cos\left(\frac{\omega T}{2}\right) \right]$$

$$f(t) = \begin{cases} 0 & t < -\frac{T}{2} \\ \frac{2A}{T} \left(t + \frac{T}{2} \right) & -\frac{T}{2} \leq t \leq 0 \\ -\frac{2A}{T} \left(t - \frac{T}{2} \right) & 0 \leq t \leq \frac{T}{2} \\ 0 & t > \frac{T}{2} \end{cases}$$



Summary

Parseval's theorem: The Fourier Series representation of a signal preserves its energy (i.e., no loss, no addition)

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)/2, \quad (5.58a)$$

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = c_0^2 + \sum_{n=1}^{\infty} c_n^2/2, \quad (5.58b)$$

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |\mathbf{x}_n|^2. \quad (5.58c)$$



Summary (cont.)

When the time period goes to infinity, the Fourier series becomes the Fourier Transform

$$\hat{\mathbf{X}}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \mathcal{F}^{-1}[\hat{\mathbf{X}}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\mathbf{X}}(\omega) e^{j\omega t} d\omega$$

$$x(t) \leftrightarrow \hat{\mathbf{X}}(\omega)$$



For the Next Class

- **Work through the textbook examples in Section 5.6 – 5.7**
- **Study Section 5-8**

How to Study:

1. **Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**